CERTIFICATE

*This is to certify that Mr./ Ms.*____

of ____________________________________ *Class, Roll No.* ___________________

Exam No. _______________________ *has satisfactorily completed his / her term work in* ____ *for the term ending* __________________ *in Year 20*___ */ 20*___.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANGA – 388 421

Date :

Sign of the Faculty *Head of the Department*

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Faculty of Technology and Engineering
Manubhai Shivabhai Patel Department of Civil Engineering

DESIGN OF STRUCTURES - II (CL 406.01)

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9		Sketch Book Drawings: (i) Detailing of Cantilever Retaining Wall (ii) Detailing of Counterfort Retaining Wall (iii) Detailing of Dog-Legged Stairways (iv) Detailing of Multistory Building Structural Elements (v) Detailing of Combined Footing (vi) Detailing of Roof Truss				

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Design of Structures - II (CL 406.01) **Date:**

TUTORIAL 1
DESIGN OF RETAINING WALL

Q-1	Design a reinforced concrete cantilever retaining wall to retain earth level with the top of wall to a height of 5.0 m above ground level. The density of soil at site is 17 kN/m^3 with a safe bearing capacity of 160 kN/m^2. Assume the angle of internal friction Ø = 30°. The coefficient of friction between soil and concrete as 0.55. Adopt M 20 grade concrete and Fe 415 HYSD bars. Also show reinforcement detail in retaining wall. Case 1: The top surface of backfill is horizontal Case 2: The top surface of backfill is inclined Case 3: Wall subjected to surcharge load of 17 kN/m2 Case 4: Soil is fully submerged into water of same height of GL
Q-2	Design a counterfort retaining wall based on the following data: Height of wall above ground level = 6 m SBC of soil = 150 kN/m^2 Angle of internal friction Ø = 30° Density of soil = 16 kN/m^3 Adopt M 20 grade concrete and Fe 415 HYSD bars.
Q-3	Design a counterfort retaining wall based on the following data: SBC of soil = 170 kN/m^2, Density of soil = 18 kN/m^3

Batches	Height of wall above ground level, m	Angle of internal friction Ø	f_{ck}, N/mm^2	f_y, N/mm^2
A1	6.5	35°	20	415
B1	7.5	33°	30	415
C1	7.5	30°	25	500
A2	6.5	27°	20	500
B2	8	30°	20	415
C2	8	35°	25	500

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TUTORIAL 2
ANALYSIS & DESIGN OF MULTI-STOREY BUILDING

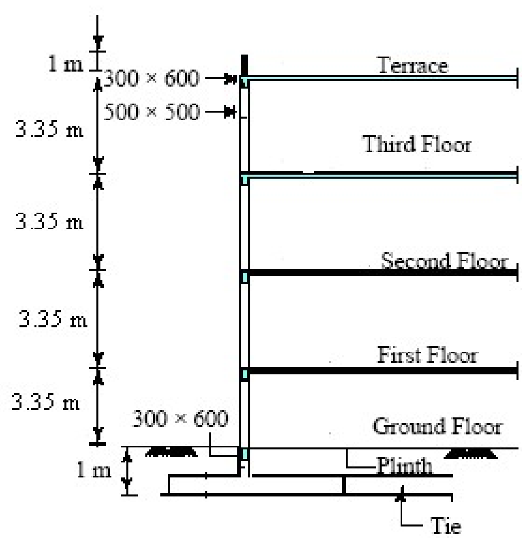
Q-1 A typical floor sectional elevation of a building is shown in figure has the following data:

1. Type of structure: Multi-storey rigid jointed frame
2. Number of storey's: 4 (G+3)
3. Floor to floor height: 3.35 m
4. Height of plinth: 1.0 m above GL
5. External walls: 250 mm thick inducing plaster
6. Internal walls: 150 mm including plaster
7. Bearing capacity of soil: 200 kN/m^2
8. Imposed loads:

Roof Level: Floor finish = 1.5 kN/m^2, Live load = 1.5 kN/m^2
Floor Level: Floor finish = 1.0 kN/m^2, Live load = 3.0 kN/m^2

9. Materials: Concrete grade M20, Steel grade Fe415
10. Unit weight of concrete = 25 kN/m^3, Unit weight of masonry = 20 kN/m^3

Design the multi-storeyed building using limit state method based on IS: 456-2000.



Typical Sectional Elevation

<u>Plan of the Building is to be submitted by group of students to the respective lab teacher.</u>

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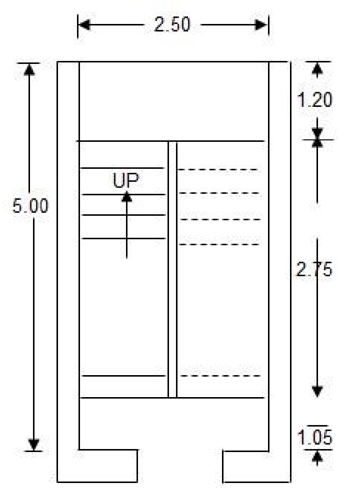
TUTORIAL 3
DESIGN OF STAIR WAYS

Q-1	Design a dog legged stair case for a residential building hall measuring 2.2 m x 4.7 m. The width of the landing is 1m. The distance between floor to floor is 3.3 m. The rise and tread may be taken as 150 mm and 270 mm respectively. The weight of floor finish is 1 kN/m^2. The materials used are M20 grade concrete and Fe415 grade steel. Sketch the details of steel. Here flight and the landing slabs spans in the same direction i.e, Flight spans longitudinally. (Batch A1)
Q-2	Design a dog legged stairs to be provided in a residential multi storied building as shown in Figure. Take M20 grade concrete and Fe415 steel. (Batch B1) 
Q-3	Design a dog legged stairs to be provided in a residential multi storied building. clear space available is 3 m x 4.8 m. Floor to floor height is 3.6 m. length of landing on either side along the direction of flight is 1.2 m. exposure condition is moderate. Take M20 grade concrete and Fe415 steel. (Batch C1)
Q-4	Design a dog legged staircase to be provided in a residential multi Storied building. Clear space available is 2.4 m x 4.8 m. Height of each flight is 1.5 m and floor height is 3.0 m. length of landing is 1.5 m and floor to floor height is 3.0 m. Length of landing on either side along the direction of flight is 1.0 m. Take Live load as 3 kN/ m^2. Take M20 grade concrete and Fe415 steel. (Batch A2)

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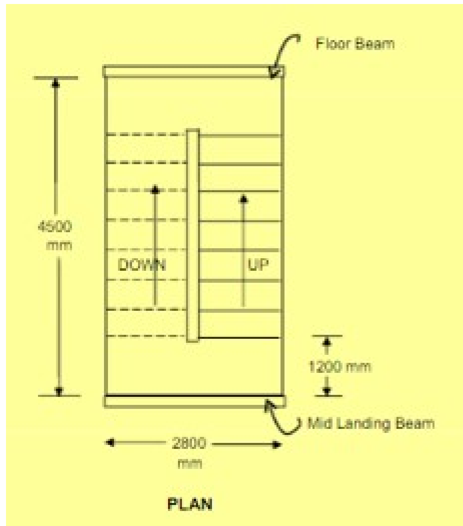
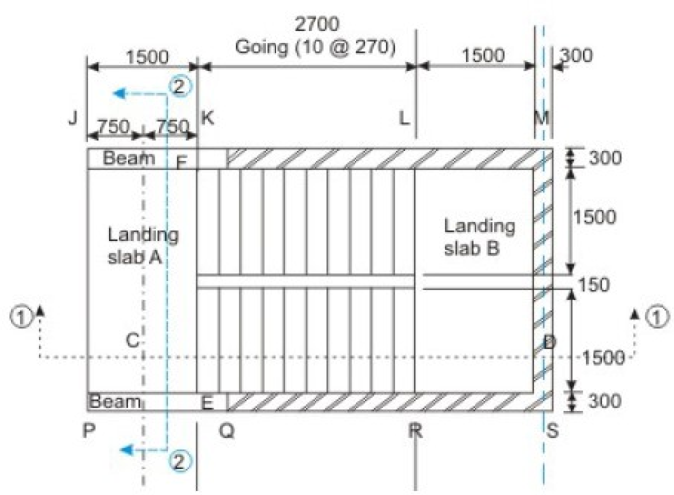
Q-5 Design a dog legged stairs to be provided in a residential multi storied building. clear space available is 2.8 m x 4.5 m. Floor to floor height is 3 m. Use riser R = 180 mm, trade T = 250 mm. Take M20 grade concrete and Fe415 steel. (Batch B2)



Q-6 Design the waist-slab type of the staircase as shown in Figure. Landing slab A is supported on beams along JK and PQ, while the waist-slab and landing slab B are spanning longitudinally as shown in Figure. The finish loads and live loads are 1 kN/m^2 and 5 kN/m^2, respectively. Use riser R = 160 mm, trade T = 270 mm. Take M20 grade concrete and Fe415 steel. (Batch C2)



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Design of Structures - II (CL 406.01) **Date:**

TUTORIAL 4
DESIGN OF COMBINED FOOTING

Q-1	Design a rectangular combined footing connecting two columns A and B, 4 m centre to centre, carrying an ultimate axial load of 1000 kN and 1400 kN respectively. The boundary line of the property is 400 mm from the outer face of the column A. Column A is 360 mm × 360 mm and column B is 420 mm × 420 mm. The bearing capacity of the soil obtained from plate load test is 106 kN/m^2. Use M20 grade concrete and Fe415 steel. Assume effective cover equal to 60 mm.

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TUTORIAL 5
DESIGN OF PLATE GIRDER

Q-1	Design a welded plate girder 20 m in span and laterally restrained throughout. It has to support a uniform load of 150 kN/m throughout the span exclusive of self-weight. Design the girder without intermediate transverse stiffeners. The steel for the flange and web plates is of grade Fe 410. Design the cross section, the end loading stiffener and connections.
Q-2	Redesign the plate girder of Q:1 using intermediate transverse stiffeners. Connections need not be designed. Use post-critical method for design.
Q-3	Design a welded plate girder of 20 m span using the tension field action for following factored forces. Mz = 5000 kNm Shear Force = 900 kN The girder is laterally restrained. Connection Need not be designed.
Q-4	Determine the flexural design strength of the following welded shape. The plate girder is simply supported and has lateral support at the end and mid span only. Consider that only flanges resist bending moment. Flanges = 300 x 200 mm Web = 1500 x 10 mm Span = 15 m

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TUTORIAL 6
DESIGN OF GANTRY GIRDER

Q-1	Design a gantry girder for the following data:

LAB	Type of crane	Crane Capaity (kN)	Max Hook Approach (m)	Span of gantry (m)	Crab Weight (kN)	Span of Crane Girder (m)	Wheel Base Width (m)	Gantry Rail Self Weight (N/m)	Diameter of Wheel (mm)	Grade of steel (MPa)	Life Span in years	Working period	Max Trip Per Hour	Working Days per year
A1	MOT	200	0.8	10	60	20	4	400	200	Fe 410	50	9:00 am to 5:00 pm	3	365
B1	MOT	300	1	9	80	18	4	400	250		100	8:00 am to 5:00 pm	4	350
C1	EOT	400	1.2	8	40	16	3.5	300	200		25	10:00 am to 8:00 pm	5	300
A2	EOT	500	1.4	7	50	14	3	300	150		50	7:00 am to 9:00 pm	6	250
B2	EOT	600	1.6	6	60	12	3	400	150		100	10:00 am to 3:00 pm	2	275
C2	EOT	800	1.8	6	40	12	2	300	250		25	11:00 am to 5:00 pm	3	250

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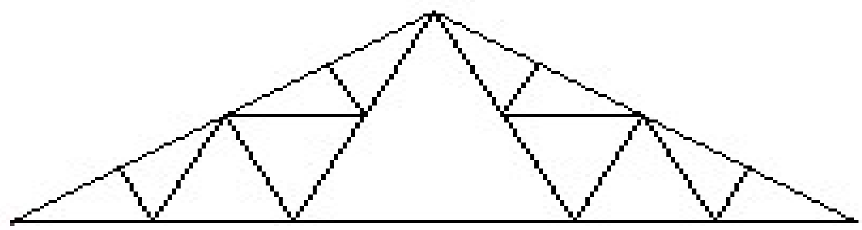
TUTORIAL 7
DESIGN OF ROOF TRUSS

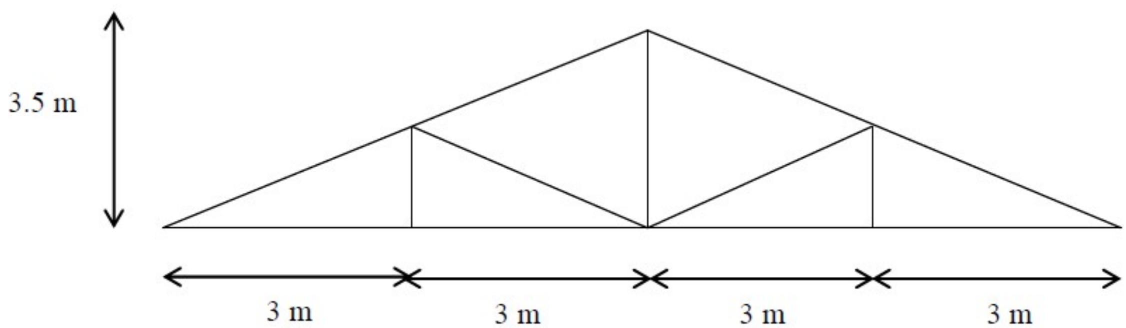
Q-1 Design and detail the roof truss and purlin for the following data:

Batch → / **Parameters** ↓	**A1**	**B1**	**C1**	**A2**	**B2**	**C2**
Truss Profile	French	Fink Fan	Trapezoidal	French	French	Fan
Permeability	Medium	Normal	Normal	Normal	Normal	Normal
Spacing	3.75 m	5 m	3.5 m	5 m	4 m	3.75 m
Span of Truss	12 m	12 m	16 m	16 m	40 m	16 m
Central Rise	3 m	3.5 m	1.75 m center, 0.5 m at end	1.6 m	3 m	3 m
Height of Truss at Eaves Level	6 m	6 m	6 m	7 m	7 m	7 m
f_y	250 MPa	250 MPa	250 MPa	250 MPa	250 MPa	250 MPa
Sheeting	Select of your choice out of G.I./A.C.					
Purlin Section	Select of your choice out of Angle, Channel, I Section					
Location	Delhi	Ahmedabad	Bangalore	Vijaywada	Vadodara	Vadodara
Connections	Select of your choice out of bolting/welding.					



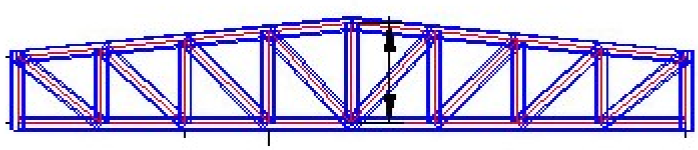
French Truss

Batch A1



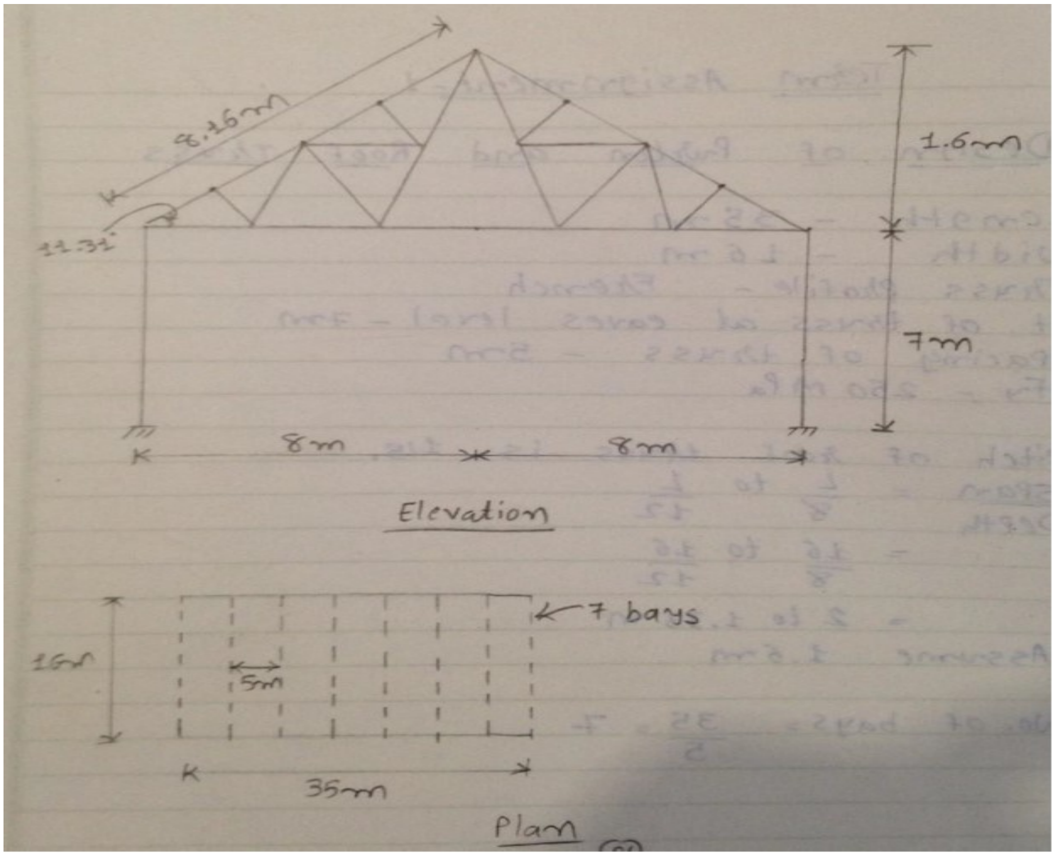
Fink Fan Truss

Batch-B1



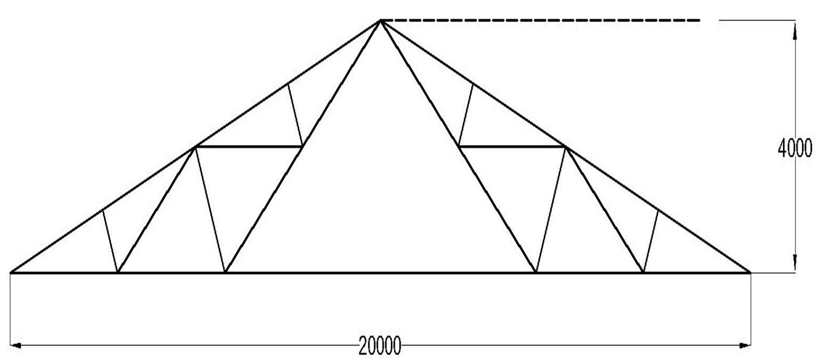
Trapezoidal Truss

Batch-C1



French Truss

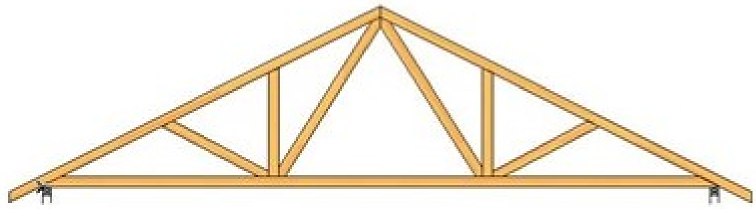
Batch- A2



ALL DIM. ARE IN MM

French Truss

Batch- B2

	 **Fan Truss** **<u>Batch - C2</u>**

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Design of Structures - II (CL 406.01) **Date:**

TUTORIAL 8
COLUMN BASES AND CAPS

Q-1	A roof truss is support over a column ISHB 300 @ 576.83 N/m and transfers following factored loads under the load combination of DL + LL and DL + WL, respectively. Compressive force P_1 = 100 kN downwards due to DL + LL Tensile force P_2 = 60 kN uplift due to DL + WL Design the column cap. Use Fe 410 grade of steel.
Q-2	Design a suitable slab base for a column section ISHB 200 @ 365.9 N/m supporting an axial load of 400 kN. The base plate is to rest on a concrete pedestal of M-20 grade. Use Fe 410 grade of steel.
Q-3	Design the bolted gusseted base for a column ISHB 450 carries an axial factored compressive force of 5500 kN. M 10 plain concrete will be provided under the base plate. Use Fe 410 grade of steel.
Q-4	A column section ISHB 350 @ 661.2 N/m carries a factored axial compressive load of 1650 kN and factored bending moment of 180 kN m. Design the base plate and its welded connection.

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